

Lab Experiments 16-17

Data Handling & Visualization: Python Code & Output

Experiment 16

Word Frequency Distribution in Customer Reviews

PYTHON

Code

```
import string
from collections import Counter

# Customer reviews dataset for a product
reviews = [
    "The product quality is amazing and delivery was fast",
    "Delivery was late but the product quality is great",
    "Great value for money, I love this product",
    "The product broke after a week, poor quality",
    "Fast delivery and amazing customer service",
    "I love the design but the quality could be better",
    "Amazing product, will definitely buy again",
    "Poor packaging led to a damaged product on delivery",
    "Great customer service and fast delivery every time",
    "The price is high for the quality offered"
]

# Preprocess: lowercase and remove punctuation
word_freq = Counter()
for review in reviews:
    review = review.lower().translate(str.maketrans('', '', string.punctuation))
    words = review.split()
    word_freq.update(words)

# Display the frequency distribution (most common first)
print("Word Frequency Distribution in Customer Reviews:")
for word, freq in word_freq.most_common():
    print(f"{word}: {freq}")
```

Output

```
Word Frequency Distribution in Customer Reviews:
the: 7
product: 6
quality: 5
delivery: 5
is: 3
amazing: 3
and: 3
fast: 3
great: 3
was: 2
but: 2
for: 2
i: 2
love: 2
a: 2
poor: 2
customer: 2
```

```
service: 2
late: 1
value: 1
money: 1
this: 1
broke: 1
after: 1
week: 1
design: 1
could: 1
be: 1
better: 1
will: 1
definitely: 1
buy: 1
again: 1
packaging: 1
led: 1
to: 1
damaged: 1
on: 1
every: 1
time: 1
price: 1
high: 1
offered: 1
```

Experiment 17

Customer Feedback Word Frequency Analysis (Top N + Bar Chart)

PYTHON

Code

```
import string
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter

# Common stop words that don't carry significant meaning
STOP_WORDS = {
    "the", "a", "an", "and", "is", "it", "of", "to", "in", "on", "for",
    "was", "were", "with", "this", "that", "i", "my", "but", "at", "as",
    "are", "be", "so", "very", "has", "have", "had", "not", "will",
    "would", "could", "can", "just", "if", "or", "we", "you", "your"
}

def load_feedback(csv_path):
    """Load the dataset from a CSV file containing a 'feedback' column."""
    df = pd.read_csv(csv_path)
    return df["feedback"].astype(str).tolist()

def preprocess(text):
    """Lowercase, strip punctuation, and remove stop words."""
    text = text.lower().translate(str.maketrans('', '', string.punctuation))
    words = text.split()
    return [w for w in words if w not in STOP_WORDS]

def compute_word_frequency(feedback_list):
```

```

    """Calculate the frequency distribution of words across all feedback entries."""
    freq = Counter()
    for entry in feedback_list:
        freq.update(preprocess(entry))
    return freq

def plot_top_n(freq, n):
    """Plot a bar graph of the top N most frequent words."""
    top_words = freq.most_common(n)
    words, counts = zip(*top_words)

    plt.figure(figsize=(8, 5))
    plt.bar(words, counts, color="darkorange")
    plt.title(f"Top {n} Most Frequent Words in Customer Feedback")
    plt.xlabel("Word")
    plt.ylabel("Frequency")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

if __name__ == "__main__":
    feedback_list = load_feedback("data.csv")
    word_freq = compute_word_frequency(feedback_list)

    n = int(input("Enter the number of top words to display: "))

    print(f"\nTop {n} Most Frequent Words:")
    for word, count in word_freq.most_common(n):
        print(f"{word}: {count}")

    plot_top_n(word_freq, n)

```

Output

```

Enter the number of top words to display: 5
Top 5 Most Frequent Words:
quality: 7
product: 6
delivery: 6
amazing: 4
fast: 4

```

Top 5 Most Frequent Words in Customer Feedback

